

Course Title	Electronic Devices & Circuits	Course Code	EE1004
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	Credit Hrs.	3+1
Knowledge Area	Electronics (KA04)	Grading Scheme	Relative
HEC Knowledge Area	Engineering Foundation	Applicable From	Spring 2023
Pre-requisite(s)	EL1001, EE1001		

Course Objective	To understand the basic structure, principles of operation, electrical characteristics and circuit models of Diodes, BJTs and MOSFETs. This course will provide foundation knowledge to analyze and design basic electronic circuits. The course will also provide basic understanding of how electronic circuits and their functions fit into larger electronic systems.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Analyze diode circuits and compute required parameters.	A1,Q1,M1	C4	2
2	Describe BJT characteristics and its operation in dc electronic circuits.	A2,Q2,M2	C2	1
3	Analyze the single stage BJT amplifier	A3,Q3,F	C4	2
4	Describe MOSFET characteristics and its operation in electronic circuits.	A4,F	C2	1

Text Books	Title	Microelectronics Circuits 6th Edition, International Version
	Author	Adel S. Sedra and Kenneth C. Smith
	Publisher	Oxford University Press, International Edition
Reference Books	Title	Electronic Devices 7th Edition
	Author	Thomas L. Floyd
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
01	Introduction, Ideal Diode, Terminal Characteristics of Diodes	2	1
02	Modeling the Diode Forward Characteristics	2	1
03	Operation in the reverse breakdown region (Zener Diodes)	2	1
04	Rectifier circuits (Half-Wave, Full-Wave, Bridge, Precision rectifiers), Limiting and clamping circuits	2	1
05	Introduction to BJTs, Current-Voltage Characteristics of BJTs	3	2
06	The BJT as an Amplifier and as a Switch, BJT circuits at DC	3	2
07	Biasing in BJT Amplifier Circuits	3	3
8	Small Signal Operation and Models	3	3
9	Basic BJT Amplifier Configurations	3	3
10	Basic BJT Amplifier Configurations	3	3
11	Discrete-Circuit BJT Amplifier	3	3
12	Introduction to MOSFETs, Device Structure & Operation	4	4
13	Current-Voltage Characteristics of MOSFETs	4	4
14	MOSFET Circuits at DC, MOSFET as an Amplifier and as a Switch	4	4
15	Biasing in MOS Amplifier Circuits, Small-Signal Operation and Models	4	4

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterms (I+II)	30.0%
Final Exam	50.0%